

Eosinophilic Pneumonia — Drug-Induced



1. NSAIDs = #1 cause

WHAT YOU SEE

Central bottle IBUPROFEN / NSAID with flame, MOST COMMON DRUG CAUSE — US

WHAT IT ENCODES

NSAIDs (ibuprofen is the prototype) are the single most common drug cause of drug-induced eosinophilic pneumonia in the US. The mechanism is a hypersensitivity reaction with eosinophil recruitment to the alveoli days to weeks after starting (or re-exposure to) the drug. Presentation overlaps with AEP/CEP — cough, dyspnea, fever, pulmonary infiltrates, often with peripheral and BAL eosinophilia.

BOARD TRAP

WRONG answer: blaming the lung disease on infection and adding antibiotics while continuing the NSAID. The discriminator is the temporal link to a culprit drug — recognizing NSAID exposure and stopping it is what reverses the disease.

2. Recent new drug

WHAT YOU SEE

Patient sash RECENT NEW DRUG

WHAT IT ENCODES

The key history is a newly started medication temporally linked to the onset of respiratory symptoms (typically within days to a few weeks). A careful medication reconciliation — including OTC drugs, antibiotics, and supplements — is essential; the diagnosis is largely clinical and rests on this temporal association plus eosinophilia and resolution after withdrawal.

BOARD TRAP

WRONG answer: anchoring on idiopathic AEP/CEP without taking a drug history. The presence of a recent new drug is exactly what defines the drug-induced category — failing to ask about medications misses a reversible, withdrawal-treated cause.

3. Nitrofurantoin step 1

WHAT YOU SEE

Bottle NITROFURANTOIN — ALSO antibiotics, STOP DRUG = STEP 1

WHAT IT ENCODES

Nitrofurantoin (classic, often after recurrent-UTI prophylaxis) and sulfonamides are textbook antibiotic causes of pulmonary eosinophilia; nitrofurantoin can cause both acute eosinophilic and chronic fibrotic lung reactions. Step 1 of management for any drug-induced eosinophilic pneumonia is stopping the offending drug.

BOARD TRAP

WRONG answer: continuing nitrofurantoin and adding a second antibiotic for presumed pneumonia. Nitrofurantoin pulmonary toxicity worsens if the drug is continued — the first and most important step is to STOP the drug, not escalate antibiotics.

4. Stop the drug first

WHAT YOU SEE

STOP sign STOP (stop the drug)

WHAT IT ENCODES

The first and most important therapeutic step is to STOP the offending drug. In mild cases, drug withdrawal alone leads to resolution within days to weeks. Removing the culprit is both diagnostic (confirms the link by resolution) and therapeutic, and it prevents recurrence on re-exposure.

BOARD TRAP

WRONG answer: jumping straight to corticosteroids while leaving the drug on board. Steroids without withdrawing the culprit treat the symptom while the cause persists — discontinue the drug FIRST; reserve steroids for severe cases.

5. Steroids only if severe

WHAT YOU SEE

Moon IF SEVERE: 1. Stop 2. Steroids PRN

WHAT IT ENCODES

Corticosteroids are added only when the respiratory illness is severe (hypoxemia, respiratory failure, or no improvement after withdrawal). The sequence is fixed: (1) stop the drug, (2) add steroids (e.g., prednisone ~0.5 mg/kg/day) if severe. Mild cases need only drug withdrawal.

BOARD TRAP

WRONG answer: giving every patient steroids regardless of severity. Mild drug-induced eosinophilic pneumonia resolves with drug withdrawal alone — steroids are reserved for severe/refractory disease, after the drug is stopped.

6. Don't continue + steroid

WHAT YOU SEE

Red X CONTINUE DRUG + ADD STEROIDS — WRONG, REMOVE FIRST

WHAT IT ENCODES

Continuing the offending drug while adding steroids is the classic WRONG management. The culprit must be removed; masking ongoing exposure with steroids risks progression and recurrence. Drug removal is the mainstay; steroids are adjunctive and severity-dependent.

BOARD TRAP

WRONG answer (explicitly): 'continue the drug and add a corticosteroid.' This is the trap distractor — never keep the offending drug on board. Remove the drug first; steroids only if severe.

7. MTX — not primary eos

WHAT YOU SEE

Crossed bottle MTX, 'can cause ILD but not primary eosinophilic — less likely'

WHAT IT ENCODES

Methotrexate causes a well-known pneumonitis/ILD (hypersensitivity-type, sometimes with peripheral eosinophilia) but it is not classically a primarily eosinophilic pneumonia — so it ranks lower as the 'eosinophilic' answer. MTX pneumonitis is managed by stopping MTX and giving steroids.

BOARD TRAP

WRONG answer: choosing methotrexate as the prototypical eosinophilic-pneumonia drug. MTX lung disease is an ILD/hypersensitivity pneumonitis pattern, not primarily eosinophilic — NSAIDs, nitrofurantoin, and sulfonamides are the higher-yield eosinophilic culprits.

8. Amiodarone — not first

WHAT YOU SEE

Crossed bottle AMIODARONE, 'pulmonary toxicity NOT eosinophilic, not first think'

WHAT IT ENCODES

Amiodarone causes chronic pulmonary toxicity (interstitial pneumonitis/fibrosis, organizing pneumonia) characterized by lipid-laden foamy macrophages on BAL — NOT an eosinophilic pattern. It is dose/duration-related; manage by stopping the drug (long half-life) plus steroids for significant disease.

BOARD TRAP

WRONG answer: picking amiodarone for drug-induced EOSINOPHILIC pneumonia. Amiodarone lung toxicity shows foamy macrophages, not eosinophils — it is the wrong answer for an eosinophilic-pattern question.

9. Allopurinol — less common

WHAT YOU SEE

Crossed bottle ALLOPURINOL, LESS COMMON

WHAT IT ENCODES

Allopurinol is a recognized but less-common cause of drug-induced eosinophilic pneumonia and can be part of DRESS (drug reaction with eosinophilia and systemic symptoms), which itself features marked eosinophilia, rash, fever, and organ involvement. Stop the drug; treat severe/systemic reactions with steroids.

BOARD TRAP

WRONG answer: ranking allopurinol above NSAIDs/nitrofurantoin as the leading drug cause. It is real but less common — and when allopurinol causes eosinophilia with rash/fever/organ injury, think DRESS rather than isolated eosinophilic pneumonia.

10. BAL eos elevated

WHAT YOU SEE

Tag BAL EOS up near patient

WHAT IT ENCODES

Diagnosis is supported by elevated BAL eosinophils (>25%) and/or peripheral eosinophilia after the drug exposure, with infiltrates on imaging. BAL also excludes infection. The clincher remains the temporal link to the drug and resolution on withdrawal.

BOARD TRAP

WRONG answer: requiring a lung biopsy to confirm. BAL eosinophilia plus the drug-exposure history and resolution after stopping the drug usually establish the diagnosis — biopsy is rarely needed.

11. Stop drug + remove machine

WHAT YOU SEE

Conveyor REMOVAL MACHINE, ANTIGEN/DRUG AVOIDANCE MAINSTAY

WHAT IT ENCODES

Drug/antigen avoidance — removing and then permanently avoiding the culprit — is the mainstay of management and prevents recurrence. Document the offending agent as an allergy/adverse reaction so it is not re-prescribed, since re-exposure can trigger a more severe relapse.

BOARD TRAP

WRONG answer: rechallenging with the drug 'to confirm' once the patient improves. Deliberate rechallenge risks a more severe reaction; avoidance is the durable management, and the agent should be flagged to prevent future exposure.

12. Add steroids if severe

WHAT YOU SEE

Moon ADD STEROIDS IF RESPIRATORY SEVERE

WHAT IT ENCODES

Reserve systemic corticosteroids (prednisone ~0.5 mg/kg/day, tapered over weeks) for respiratory-severe cases — significant hypoxemia, respiratory failure, or failure to improve after drug withdrawal. Mild cases recover on withdrawal alone, so steroids are severity-driven, not automatic.

BOARD TRAP

WRONG answer: starting high-dose steroids before/instead of stopping the drug. The order is fixed — withdraw the drug first; steroids are added only for severe respiratory compromise.

13. Acute HP mimic

WHAT YOU SEE

Panel ACUTE HP — BAL lymphocytes not eos

WHAT IT ENCODES

Differential: acute hypersensitivity pneumonitis (HP) follows an inhaled antigen and shows BAL LYMPHOCYTOSIS (CD8-predominant), not eosinophilia. Both are exposure-related hypersensitivity reactions, so the BAL cell differential separates them.

BOARD TRAP

WRONG answer: calling drug-induced eosinophilic pneumonia 'HP.' HP BAL is lymphocytic; drug-induced eosinophilic pneumonia has BAL eosinophilia — the lavage cell type is the discriminator.

14. AEP mimic

WHAT YOU SEE

Panel AEP — no drug exposure

WHAT IT ENCODES

Differential: idiopathic acute eosinophilic pneumonia (AEP) looks identical (acute febrile illness, BAL eosinophilia, steroid-responsive) but by definition lacks a drug trigger and is linked instead to new smoking/vaping or dust. Here there IS a culprit drug, which categorizes it as drug-induced.

BOARD TRAP

WRONG answer: labeling a clearly drug-triggered case as idiopathic AEP. The presence of a temporally related offending drug moves it out of the idiopathic AEP category — management adds drug withdrawal, not just steroids.

15. Hypereosinophilic syndrome

WHAT YOU SEE

Panel HYPEREOSINOPHILIC SYNDROME — organs, >1500 eos

WHAT IT ENCODES

Differential: hypereosinophilic syndrome (HES) — persistent peripheral eosinophilia >1500 cells/ μ L for $\geq$ 6 months with multiorgan damage (heart/endomyocardial fibrosis, skin, GI, neuro). Unlike a single-organ drug reaction, HES is systemic and persistent; treat with steroids and, if FIP1L1-PDGFR α -positive, imatinib.

BOARD TRAP

WRONG answer: diagnosing HES off a single episode of drug-related pulmonary eosinophilia. HES requires persistent eosinophilia >1500 with multiorgan involvement — a transient, drug-linked, lung-limited process that resolves on withdrawal is drug-induced eosinophilic pneumonia, not HES.

16. Larva migrans

WHAT YOU SEE

Panel LARVA MIGRANS — travel history + parasite

WHAT IT ENCODES

Differential: parasitic eosinophilic lung disease (Löffler syndrome / larva migrans) from transpulmonary migration of helminths — Ascaris, Strongyloides, hookworm — causing transient migratory infiltrates with eosinophilia. Ask about travel, soil/exposure, and check stool O&P and Strongyloides serology, especially before giving steroids (which can precipitate Strongyloides hyperinfection).

BOARD TRAP

WRONG answer: giving corticosteroids for presumed drug-induced disease without excluding Strongyloides in an at-risk patient. Steroids can trigger fatal Strongyloides hyperinfection/disseminated strongyloidiasis — screen and, if indicated, treat with ivermectin first.

17. Drug = #1 of all eos infiltrates

WHAT YOU SEE

SCOREBOARD: US #1 cause of pulmonary eos infiltrates = drug-induced

WHAT IT ENCODES

Board pearl: drug-induced disease is the leading identifiable cause of pulmonary eosinophilic infiltrates in the US. The practical takeaway in any eosinophilic-pneumonia vignette is to scrutinize the medication list first — drugs are common and the cause is reversible by withdrawal.

BOARD TRAP

WRONG answer: defaulting to an idiopathic or parasitic cause without reviewing medications. In the US the most common identifiable cause is drugs — the medication history is the highest-yield first step and points to a reversible diagnosis.